BRIAN MUHS

Senior Software Developer | AI-Integrated B2B SaaS

Scranton, PA briankmuhs@hotmail.com +1 570 507 8657

SUMMARY

Senior Software Developer with 13+ years of experience designing large-scale platform systems, specializing in AI-integrated learning and service discovery architectures. Proven expertise in building high-performance backend ecosystems using Java, Spring-based microservices, React.js, Next.js, Angular, Python and modern cloud-native technologies across Amazon Web Services. Focused on delivering scalable SaaS platforms, distributed processing pipelines, and automation-driven engineering solutions that power production-grade digital products and enterprise workflows.

EXPERIENCE

Larson Design Group — *Senior Software Developer (AI Integrated SMM Research Platform)*

USA | Remote | Full-time | Mar 2024 –

- Led development of a scalable social media marketing service research platform using Java-based backend architecture, supporting aggregation and management of millions of provider service records from global marketing service suppliers.
- Architected microservices-oriented backend systems using the Spring ecosystem, separating provider aggregation, search indexing, order lifecycle processing, billing operations, and AI normalization services into independently deployable components.
- Developed RESTful APIs using Spring Boot and Spring MVC following domain-driven architectural principles, maintaining clear separation across controller, business logic, and persistence layers.
- Implemented transactional persistence using Hibernate with MySQL, optimizing relational mappings, indexing strategies, and query execution performance for large-scale service catalog operations.
- Built a high-performance discovery engine using Elasticsearch integrated with backend services, enabling advanced multi-attribute filtering, semantic relevance ranking, and search optimization that reduced average query latency from ~1.8s to under 400ms.
- Integrated AI/LLM-based normalization pipelines to standardize heterogeneous provider metadata and improve semantic search accuracy, contributing to approximately 19% improvement in successful order conversions.
- Implemented event-driven communication and data streaming pipelines using Apache Kafka to support reliable inter-service message exchange for order lifecycle updates, provider synchronization events, and AI processing triggers, improving system decoupling and scalability.
- Designed distributed caching mechanisms using Redis for query result caching, session optimization, and synchronization locking, reducing database read load by ~41% and improving API response latency by ~33% during peak traffic.
- Implemented secure authentication and authorization using Spring Security with JWT-based identity management and role-based access control across multi-tenant environments.
- Integrated payment processing services including Stripe and Cryptomus by enforcing webhook signature validation, idempotent transaction handling, and fault-tolerant financial event processing.
- Built automated backend testing pipelines using JUnit and Spring Test frameworks for unit, integration, and API-level validation to maintain production stability.
- Developed responsive enterprise dashboards and service management interfaces using Angular, implementing modular component architecture, reactive forms, and lazy-loaded routing strategies for performance optimization.
- Utilized TypeScript, RxJS reactive programming patterns, and state synchronization techniques to manage complex UI workflows including search filtering, analytics visualization, and multi-provider order management.
- Built UI styling systems using SCSS and component-level design abstraction to maintain consistency across large-scale enterprise interfaces.

- Developed Python-based automation utilities using Python and Playwright for provider intelligence extraction, including pricing signal collection, execution benchmarking, and reliability scoring with anti-bot protection and parallelized execution.
- Deployed backend and frontend services using containerized CI/CD pipelines, supporting high availability releases, monitoring integration, and operational runtime stability.
- Designed, deployed, and maintained platform infrastructure on AWS, leveraging EC2-based compute environments, S3 object storage for application and AI-generated assets, and containerized CI/CD deployment workflows to support scalable and highly available service operations.
- Contributed to architecture discussions, performance audits, and code reviews in an Agile team environment.

Tech: *Java, Spring Boot, Spring Framework, Spring Security, Hibernate/JPA, MySQL, Elasticsearch, Redis, Kafka, RESTful Microservices Architecture, JWT Authentication, Angular, TypeScript, RxJS, SCSS, AI/LLM Backend Integration, Payment Gateway Integration, Docker, Linux, CI/CD, JUnit, Python Automation, Playwright*

Discovery Machine® Inc — *Senior Software Engineer (AI Integrated LMS Platform)*

USA | On-site | Full-time | Oct 2017 – Mar 2024

- Designed and evolved core platform services using Java with the Spring ecosystem, transitioning critical workflows from tightly coupled implementations toward modular, independently deployable service boundaries.
- Built high-throughput RESTful APIs using Spring Boot, Spring MVC, and Spring Security, enforcing domain-driven design principles, structured error handling, and layered service abstractions.
- Implemented authentication and authorization infrastructure using JWT-based identity tokens integrated with Spring Security, supporting multi-role access control across creator, learner, and administrative domains.
- Developed complex persistence models using JPA with PostgreSQL, supporting hierarchical course structures, enrollment state machines, certification issuance pipelines, and subscription lifecycle transactions.
- Applied database performance optimization techniques including composite indexing, connection pool tuning, and ORM query refinement, contributing to approximately ~24% improvement in query execution efficiency for high-frequency workloads.
- Introduced Redis-based distributed caching strategies for session metadata and frequently accessed platform resources, reducing database read load by roughly ~48% under peak concurrency scenarios.
- Engineered asynchronous backend processing pipelines for analytics aggregation, email automation, content transformation tasks, and AI inference triggers.
- Implemented event-driven communication across platform subsystems using Apache Kafka for reliable message streaming related to enrollment events, assessment results, and learning activity telemetry.
- Integrated large language model APIs into backend services to support automated course outline generation, quiz suggestion recommendations, instructional summarization, and semantic metadata enrichment, improving creator workflow efficiency by approximately ~18%.
- Designed personalization logic leveraging behavioral engagement metrics, course completion signals, and interaction history to generate adaptive learning recommendations.
- Developed enterprise-grade user interfaces using React (JavaScript library) and Next.js with hybrid server-side rendering and static generation strategies to improve initial render performance and SEO accessibility.
- Built scalable component-driven UI architecture using TypeScript, custom hooks abstraction layers, and reactive state orchestration patterns.
- Implemented advanced frontend performance optimization including dynamic module imports, route-level code splitting, memoized rendering pipelines, and virtualized list handling for large dataset views.
- Designed reusable design system components for dashboards, course authoring interfaces, analytics panels, and monetization workflow pages.
- Utilized modern frontend build tooling including Webpack optimization configurations for efficient asset bundling and runtime execution.
- Built secure educational video delivery architecture using HTTP Live Streaming (HLS) with tokenized access validation and storage-backed media pipelines.

- Implemented signed resource access mechanisms to protect premium learning content while maintaining low-latency playback across distributed users.
- Architected and operated cloud-native platform infrastructure on Amazon Web Services supporting compute orchestration, object storage, and service runtime execution.
- Provisioned and automated infrastructure deployments using Terraform for reproducible environment lifecycle management.
- Orchestrated containerized microservices workloads using Kubernetes clusters for service scaling, health recovery, and workload scheduling.
- Implemented platform monitoring and observability pipelines using Datadog for distributed tracing, application performance metrics, and system reliability tracking.
- Established CI/CD automation using Docker-based pipelines to support controlled release deployment, rollback safety, and staging-to-production promotion workflows.
- Participated in platform architecture planning, technical design reviews, and Agile iterative delivery cycles.

Tech: *Java, Spring Boot, Spring Framework, Spring Security, Hibernate/JPA, MySQL, PostgreSQL, Redis, Apache Kafka, React.js, Next.js, TypeScript, SCSS/TailwindCSS, SCORM, xAPI (TinCan API), AI/LLM Integration, Terraform, Kubernetes, Datadog, AWS, Docker, CI/CD, Git*

Scientech — *Software Engineer (Clinical Decision Support System)*

USA | On-site | Full-time | Oct 2014 – Oct 2017

- Contributed to the design and stabilization of a Clinical Decision Support System (CDSS) used for drug interaction checks, contraindication alerts, and treatment validation within clinical workflows.
- Modeled medical ontologies such as RxNorm and MedRT in Neo4j, defining graph schemas, node labels, and relationship mappings to support accurate semantic traversal and drug interaction inference.
- Optimized Cypher queries by restructuring traversal paths, improving relationship indexing, and using query profiling tools, resulting in ~30% faster execution for core clinical decision logic.
- Developed backend services in Python using FastAPI, implementing structured routing, Pydantic-based request validation, consistent error handling, and clean service-layer abstractions.
- Built asynchronous processing flows using asyncio to handle concurrent clinical event evaluation and alert generation without blocking API performance.
- Integrated PostgreSQL for transactional data including user records, system configurations, and audit logs, ensuring consistency between relational and graph-based components.
- Implemented event-driven communication components using Apache Kafka for reliable clinical event streaming, decoupling decision computation modules from ingestion and notification subsystems.
- I Implemented OAuth2 and JWT-based authentication with role-based access control (RBAC) to secure administrative and clinical-facing endpoints.
- Developed internal administrative dashboards using React.js and Ant Design, including ontology management interfaces, alert review panels, and system configuration tools.
- Wrote unit and integration tests using pytest to validate API endpoints and clinical rule execution logic, reducing regression issues during ontology updates.
- Assisted with Docker-based containerization and CI/CD deployment workflows, supporting environment consistency and smoother production releases.

Tech: *Python, FastAPI, Neo4j, CQL, CDSS, RxNORM, MedRT, Asyncio, Concurrency, Graph Databases, React.js, Antd, pytest, asyncio, pydantic, Apache Kafka*

Landauer Games — *Software Engineer*

USA | Remote | Contract | May 2012 – Oct 2014

- Designed and developed backend services for game platform functionality, focusing on scalable server-side APIs, session management workflows, and persistent gameplay data processing.
- Built RESTful microservice-style backend components using Java with Spring and Spring Boot, implementing layered architecture separation across controllers, business logic, and data access services.
- Implemented object-relational mapping and transactional persistence layers using Hibernate/JPA, designing normalized entity models and optimizing SQL execution paths for platform gameplay and user state storage.

- Developed JSON-based communication protocols for client-server interaction, supporting lightweight data serialization and high-throughput request handling for gaming service endpoints.
- Engineered authentication and session management modules using token-based security patterns, including secure user identity validation and permission enforcement.
- Enhanced backend reliability by writing comprehensive unit and integration tests using JUnit and mock service testing strategies.
- Worked with relational database systems such as MySQL for structured gameplay data storage, indexing strategy design, and query performance tuning.
- Implemented asynchronous processing components where applicable to improve response latency for gameplay event handling and background computations.
- Supported build automation, version control workflows, and deployment preparation pipelines within Java-based backend environments.
- Participated in debugging, feature extension, and performance stabilization efforts for platform service modules.

Tech: *Java, Spring Boot, Spring Framework, Hibernate, Microservice architecture, Restful Backend Architecture, JSON, MySQL, Junit*

SKILLS

Front End: AngularJS, Vue.js, React, Next.js, Nuxt, Livewire, React Native; TypeScript; HTML5/CSS3; Tailwind CSS, SCSS; WebSockets

Back End: Microservice architecture, Java(Spring Boot/MVC/Security), Hibernate, PHP(Laravel, Symfony), Node.js (NestJS/Express), Python (FastAPI/Flask/Django), REST & GraphQL APIs; Background jobs, queues, rate-limiting, auth (JWT, OAuth2); Kafka

Data/Storage: PostgreSQL, MySQL; MongoDB/DynamoDB; Neo4j; Vector DBs: FAISS, Pinecone, Qdrant, OpenSearch; ElasticSearch; Redis

AI/ML: LLMs (GPT-4o/Claude/Mistral/Llama), RAG, LangChain, LangGraph, Prompt Engineering

Cloud/DevOps: GCP, AWS, Azure; Docker; CI/CD; Git

Domain & Specialized Systems: SCORM, xAPI (TinCan API), Medical Ontologies (RxNORM, MedRT)

EDUCATION

Penn State University — *Bachelor of Science – BS. Computer Science*

2008 – 2012